Quiz 4: October 23

Name: _____

SID: _____

Problem 1. Suppose that $S, T : V \to V$ are linear maps such that $TS = ST$. If U is an invariant subspace of T, show that

$$\ker(S) = \{v \in V : Sv = 0\}$$

is an invariant subspace of T as well. **Solution:** Let $v \in \ker(S)$ be arbitrary, so $Sv = 0$. Then, consider

that

$$S(Tv) = T(Sv) = T(0) = 0$$

since T is linear, which implies $Tv \in \ker(S)$. This shows $T(\ker(S)) \subseteq \ker(S)$ as desired.

Problem 2. Let $T : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ be the linear function given by

$$T(p(x)) = 2xp'(x).$$

Find the eigenvalues of T.

Solution: Under the basis $\{1, x, x^2\}$, we have the matrix for T

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Since this matrix is diagonal (or upper triangular), we may read off the eigenvalues $0, 2, 4$.